TECHNICAL WHITEPAPER & SYSTEM SPECIFICATION

Scout Screener

Autonomous Recruiter Architecture, Evidence-Linked Scoring & Multi-Role Matching Framework

Version: 2.4.0 (Production)
Platform: scout-screener.vercel.app
Date: November 2026
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1. Executive Summary & Design Philosophy

Scout Screener is a next-generation talent evaluation and autonomous recruitment intelligence system engineered to address the critical flaws of legacy Applicant Tracking Systems (ATS). Conventional recruitment automation relies on opaque keyword matching or black-box classifiers that discard qualified talent without auditability. Scout Screener implements a fundamental architectural shift: **Deterministic, Evidence-Based Human-in-the-Loop Screening**.

The Three Core Tenets of the Scout Framework:

- **"Not on the CV" means Verify, Never Reject:** Absence of explicit keywords in an applicant's resume is treated as a conversational gap to be verified via interactive chat, never an automatic disqualification.
- **Strict Evidence Attribution (Zero Black-Box):** Every percentage point in an applicant's match score is mathematically backed by direct, verbatim quotes extracted from their source CV text.
- **Human Decision Primacy:** Scout acts as an autonomous analytical co-pilot. It synthesizes signals, structures evidence, and flags discrepancies, but final progression, shortlisting, and hiring decisions remain 100% human-controlled.

2. High-Level System Architecture

The system operates as an edge-optimized, privacy-preserving hybrid platform comprising four decoupled tiers:

Client Tier (Edge Browser)
Zero-dependency Vanilla SPA • Local-first IndexedDB / LocalStorage • Client-side PDF stream extraction • In-browser Tesseract OCR worker

Edge Serverless API (Vercel)
Node.js Serverless runtime • Payload encryption • Web search & DuckDuckGo scraping proxy • Stateless request routing

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Inference Engine (Groq LPUs)
Llama-3.3-70b-versatile • Deterministic temperature (0.1) • Strict JSON Schema enforcement • 800 tokens/sec ultra-low latency

Persistence Tier (Dual-Mode)
Upstash Redis REST KV store • PostgreSQL relational backend • Zero-setup offline local fallback

System Topology Matrix

Component	Technology	Responsibilities & Failure Handlers
Frontend Shell	HTML5, CSS3, Vanilla JS	Responsive mobile-first employer dashboard, real-time reactive hash router, drag-and-drop batch flyer ingestion, Kanban pipeline.
Document Ingestion	PDF.js, Tesseract.js OCR	Client-side extraction of text from PDF documents, image flyers, and screenshots. Runs in background Web Workers without backend data transfer.
Scoring Engine	`engine.js`	Deterministic weighted evaluation (Must-Haves 3x, Nice-to-Haves 1x), match status classification, evidence slicing, signature hashing.
Web Enrichment	`api/web-enrich.js`	Internet verification of job legitimacy, scraping official career portals and contacts, filtering fraudulent postings.
Recruiter Co-Pilot	`hire.js` ("Ask Scout")	Natural Language Intent Parser mapping recruiter queries ("show me the top 30") into interactive filtered candidate decks.

3. Deep-Dive: Core Algorithms & Pipelines

3.1. The Mathematical Scoring Model (`Engine.score`)

Rather than subjective text summaries, Scout calculates an objective, reproducible compatibility score across all extracted requirements:

$$\text{Overall Match (\%)} = \lfloor \left(\sum (w_i \times m_i) / \sum w_i \right) \times 100 \rfloor$$

Where for each requirement i:

- **Weight (w_i):** Assigned as `must` (Weight = 3) or `nice` (Weight = 1). Must-have requirements carry triple the mathematical influence on the final ranking.
- **Match Coefficient (m_i):**
  - `1.0 (Met):` Directly substantiated by quoted text in the candidate's CV or verified portfolio.
  - `0.5 (Verify / Partial):` Inconclusive, partially stated, or self-reported in an interview without corroborating work history.
  - `0.0 (Missing):` Explicitly absent from the CV and application records.
- **Verbatim Evidence Pinning:** Each matched requirement stores `quote` (exact substring from CV) and `source` reference, allowing recruiters to click any score bar segment to view the exact CV sentence.

3.2. Autonomous Web Search & Verification Pipeline

When a job is parsed from a flyer screenshot or raw text, the system invokes `/api/web-enrich` to cross-reference the live web:

1. **Query Generation:** Formulates targeted web search parameters combining job title, hiring company name, and location.
2. **DuckDuckGo Scraping & Filtering:** Retrieves real-time search results, career listing indices, and domains.
3. **AI Synthesis & Verification:** Evaluates search results to isolate official employer domains (e.g. `flutterwave.com/careers`), preventing job seekers from applying through unauthorized third-party boards.
4. **Contact & Metadata Enrichment:** Extracts official recruiting email addresses, recruiter WhatsApp contacts, and verified salary ranges.

3.3. Batch Multi-Flyer Ingestion Engine

Recruiters frequently receive job advertisements as graphic posters, flyers, or social media snapshots. Scout accepts simultaneous uploads of up to 20 flyer images in a single drag-and-drop operation:

- **Parallel Worker Processing:** Dispatches each image to an isolated in-browser Tesseract OCR worker.

- **Intelligent Noise Filtering:** Eliminates flyer slogans, non-essential contact noise, and visual artifacts to isolate core role competencies.
- **Automatic Pipeline Provisioning:** Instantly generates dedicated job records, application links, and requirement sets for every flyer uploaded.

3.4. Cross-Job Talent Match Matrix ("See Who Fits Where")

Traditional recruiting evaluates candidates in rigid silos. Scout's **Talent Matrix** algorithm evaluates the entire candidate pool across all active organization openings simultaneously ($C imes J$ evaluation matrix):

Applicant Name	Applied Role	Product Designer (Fit)	Founding Backend (Fit)	Data Analyst (Fit)	Recommended Action
Amaka O.	Backend Engineer	42%	**86% (Top Match)**	68%	Fast-track Backend Pipeline
Ibrahim K.	Backend Engineer	35%	64%	**84% (High Fit)**	*Cross-offer to Data Analyst role*
Chioma E.	UI/UX Designer	**92% (Top Match)**	20%	45%	Shortlist for Product Designer

4. Recruiter Co-Pilot ("Ask Scout") & Conversational Workflows

Accessible via the floating action co-pilot button located in the employer interface, **Ask Scout** translates natural language recruiter directives into high-speed pipeline actions:

Supported Recruiter Queries:

- `"Show me the top 30"` → Computes weighted scores for all applicants, ranks them in descending order, and displays rich candidate cards.
- `"Above 80% in Lagos"` → Applies multi-dimensional filtering across geographic and competency criteria.
- `"Who fits which job?"` → Launches the cross-role talent matrix.
- `"Who completed the chat?"` → Filters for candidates who completed the short-chat interview.

Interactive Card Action Deck:

- **Match Percentage Badge:** Color-coded visual indicator (≥80% green, 60-79% blue, <60% amber).
- **Requirement Micro-Bar:** Visual segmentation of must-haves vs nice-to-haves.
- **One-Click Drilldown:** Tapping any card opens the candidate's verified CV, contact details, interview responses, and line-by-line evidence links.

5. Security, Privacy & Ethical AI Governance

Scout is architected with strict compliance, data sovereignty, and anti-bias principles:

1. Blind PII Screening

Applicant telephone numbers and email addresses are cryptographically withheld from recruiter view until the recruiter takes an explicit action to reveal them, minimizing unconscious bias during initial evaluation.

2. Capability URL Isolation

Recruiter administration routes require a high-entropy capability key (`?key=...`). No global admin accounts exist with backdoor access to private company applicant pools.

3. 100% Auditability

Every AI extraction and scoring outcome includes a full reasoning trace and prompt transcript, ensuring compliance with international labor laws and automated hiring regulations.

6. Summary & Platform Capabilities At-a-Glance

Feature	Traditional ATS / Job Boards	Scout Screener Framework
Screening Methodology	Opaque keyword frequency filters	Semantic requirement matching with quoted proof
Candidate Input	Rigid, tedious forms (often 30+ minutes)	1-Click PDF upload, phone screenshot, or pasted text
Recruiter Exploration	Manual spreadsheet clicking	Conversational Assistant ("Show me the top 30")
Cross-Role Discovery	None (candidates rejected per job)	Dynamic $N imes M$ Cross-Job Talent Matrix
Job Authenticity	Unchecked job flyer spam	Automated web scraping & official domain verification